DESIGN DOCUMENTATION

MindBridge

Multimodal Intelligence Learning Platform

A web application that presents middle-school curriculum concepts through configurable learning pathways and evaluates learning outcomes.

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1. Introduction

1.1 Purpose

This document gives the design of the MindBridge Multiimodal Intelligence Learning Platform. The High Level Design (HLD) shows the architecture and components. The Low Level Design (LLD) shows the classes, database, APIs and algorithms needed to code it.

1.2 Goal

- Ask the student a questionnaire to estimate learning preferences across the eight approved intelligence types.
- Calculate preference scores from self-report and scenario choices; keep response time separate.
- Build an indicative learning-preference profile and identify primary and secondary preferences; do not label these as fixed intelligence types.
- Give the preference profile to Module 2 as an input for recommending content representations.
- Finish in 20 minutes: 24 questions: 8 self-report + 16 scenario based, each with its own time limit

1.3 Design Decisions Made:

These are configurable implementation decisions where the requirements do not specify a value. Provisional values must be reviewed during a student pilot; they are not validated measurements.

No.	Decision	Reason
D1	24 questions: 8 self-report questions and 16 scenario-based questions.	Provisional assessment structure for Classes 6–8. Question mappings must provide reasonable coverage across all eight supported learning pathways and be evaluated during pilot testing.
D2	Each scenario question option maps to one or more intelligence types.	Scenario options represent different ways to explore the same concept; options are preference choices, not right or wrong answers.
D3	Self-report uses a 5-point scale; response weight = (rating - 1) / 4	Gives a value between 0 and 1
D4	No time penalty is applied to preference scores	Avoid penalizing students for reading speed or response latency. Response time is recorded separately and may be used only as a secondary factor when resolving tied or near-tied preference scores.
D5	The assessment targets a total duration of 20–30 minutes. Individual question time limits, if required, shall be configurable.	Provisional duration. A question not answered in time is recorded as timed out and the test moves on; timing must be pilot-tested.
D6	Time is measured on the server	Refresh or tampering cannot reset the timer (PRE-N3, PRE-T4)
D7	Teacher/parent observation is optional and separate from the student preference score	Avoid silently mixing observer judgments with student self-reports; enable only after permissions and consent are approved.

Table 1: Design decisions Table

2. High Level Design (HLD)

2.1 System Architecture

The module uses a 4-layer architecture: Presentation, API, Service and Data. Each layer talks only to the layer below it.

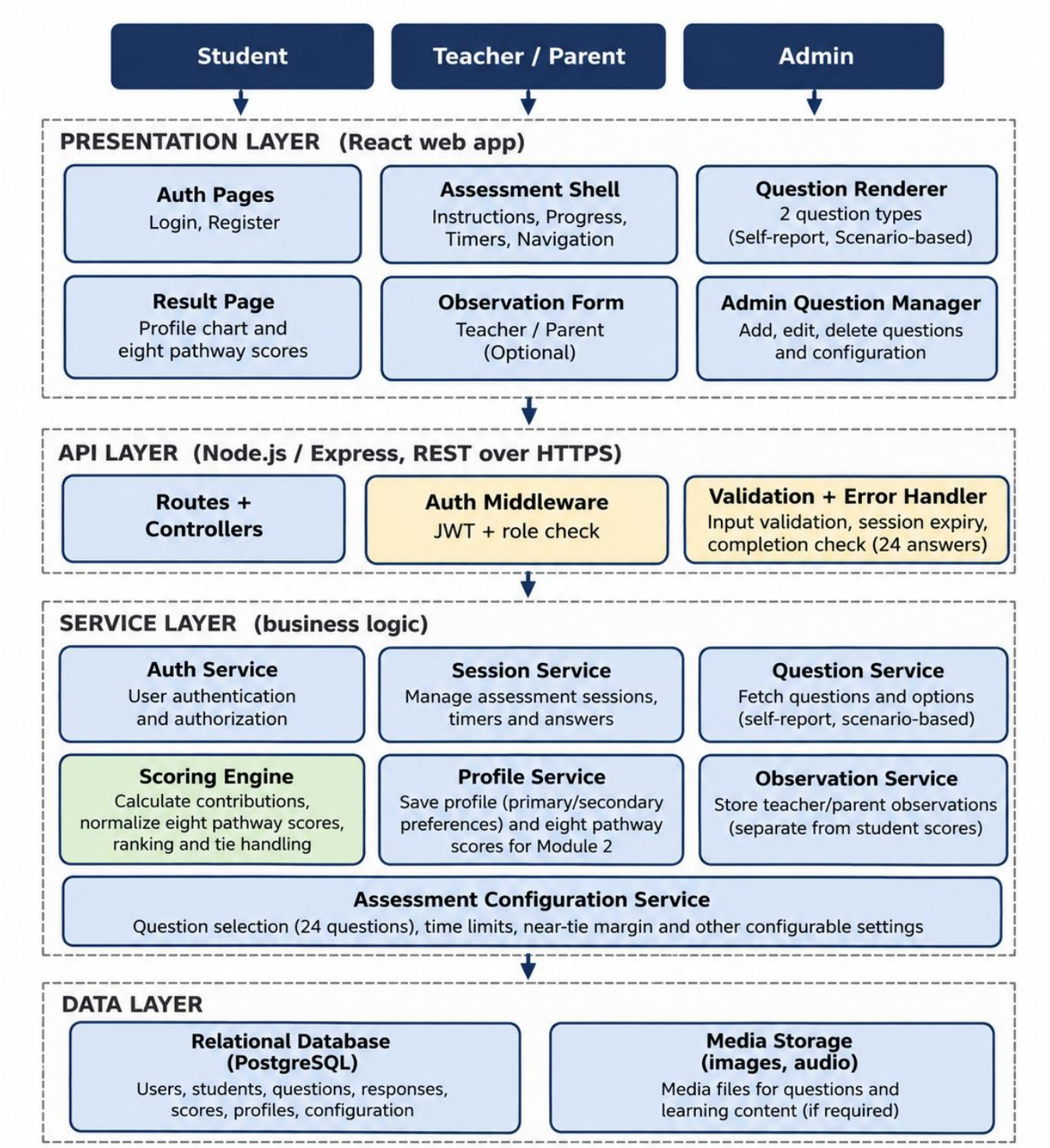


Figure 1: HLD architecture of the Pre-Assessment module

2.2 Component Overview

Layer	Component	Responsibility
Presentation	Auth UI	Registration and login
Presentation	Assessment Shell	Instructions, progress bar, countdown timers (question and total), navigation
Presentation	Question Renderer	Shows approved question types: self-report and scenario based.
Presentation	Result Page	Shows indicative learning-preference scores and primary/secondary preferences.
Presentation	Observation Form	Optional teacher observation form; only for authorized users if approved.
Presentation	Admin Question Manager	Add, edit, delete questions and weights
API	Controllers and routes	Receive requests and return responses (REST, JSON)
API	Auth middleware	Checks JWT token and user role
API	Validation and error handler	Checks input and gives clear error messages
Service	Auth Service	Register, login, password hashing
Service	Session Service	Start or resume session, serve next question, save answers, enforce time limits
Service	Question Service	Select balanced question set; admin CRUD
Service	Scoring Engine	Applies formula, normalises, ranks
Service	Profile Service	Saves the preference profile and provides primary/secondary preferences to Module 2.
Service	Observation Service	Stores optional observations separately from student preference scores.
Service	Config Service	Reads versioned question weights and timing limits; no preference-score time penalty.
Data	Relational database	Users, questions, sessions, responses, scores, profiles
Data	Media storage	Images and audio used in tasks

Table 2: Components overview Table

2.3 Main Data Flow

The sequence below shows one full preference assessment, from start to result. Steps 5 to 10 repeat for each of the 24 questions.

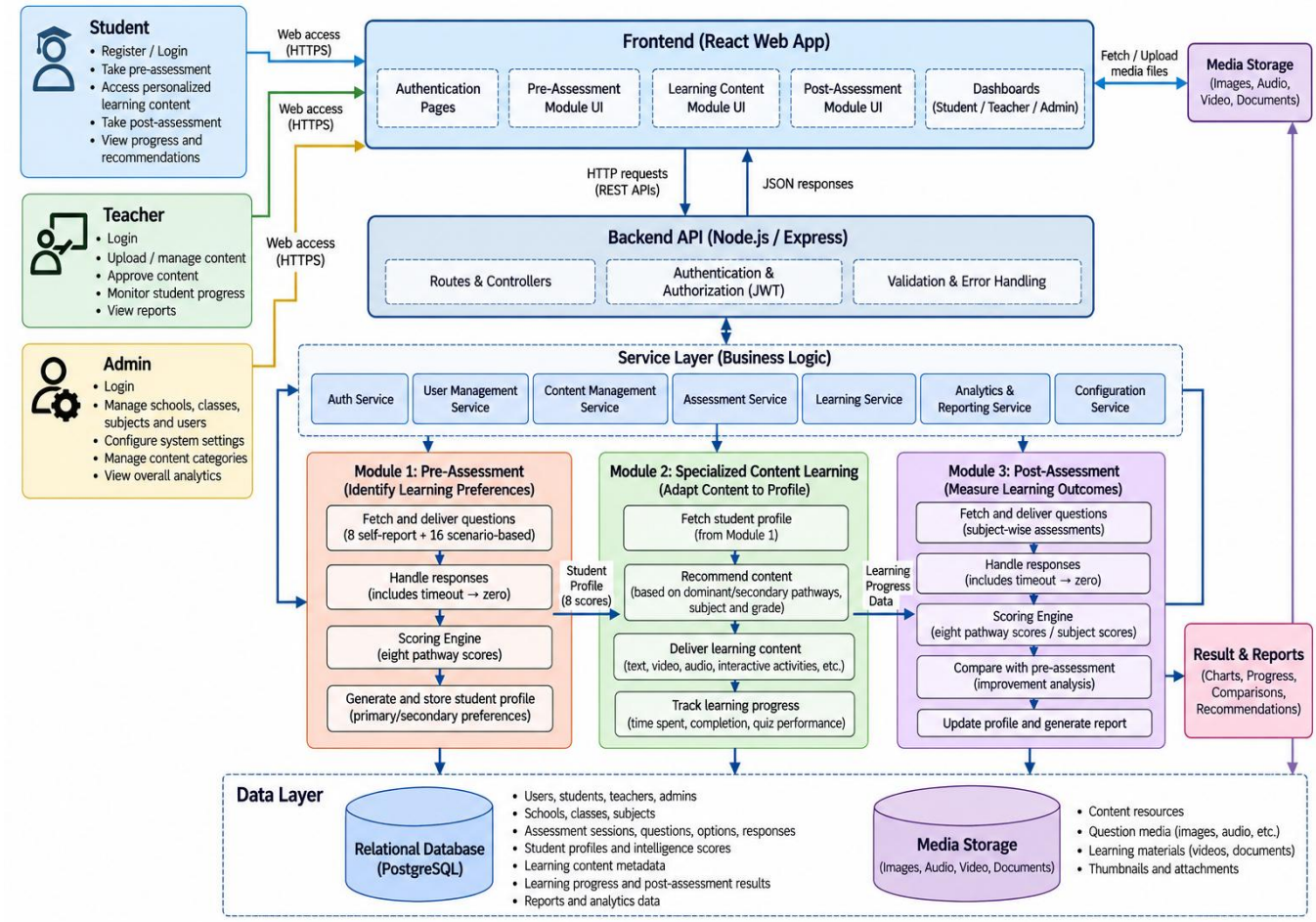


Figure 2: Main data flow of MindBridge

2.3.1 Data flow of pre-assessment module:

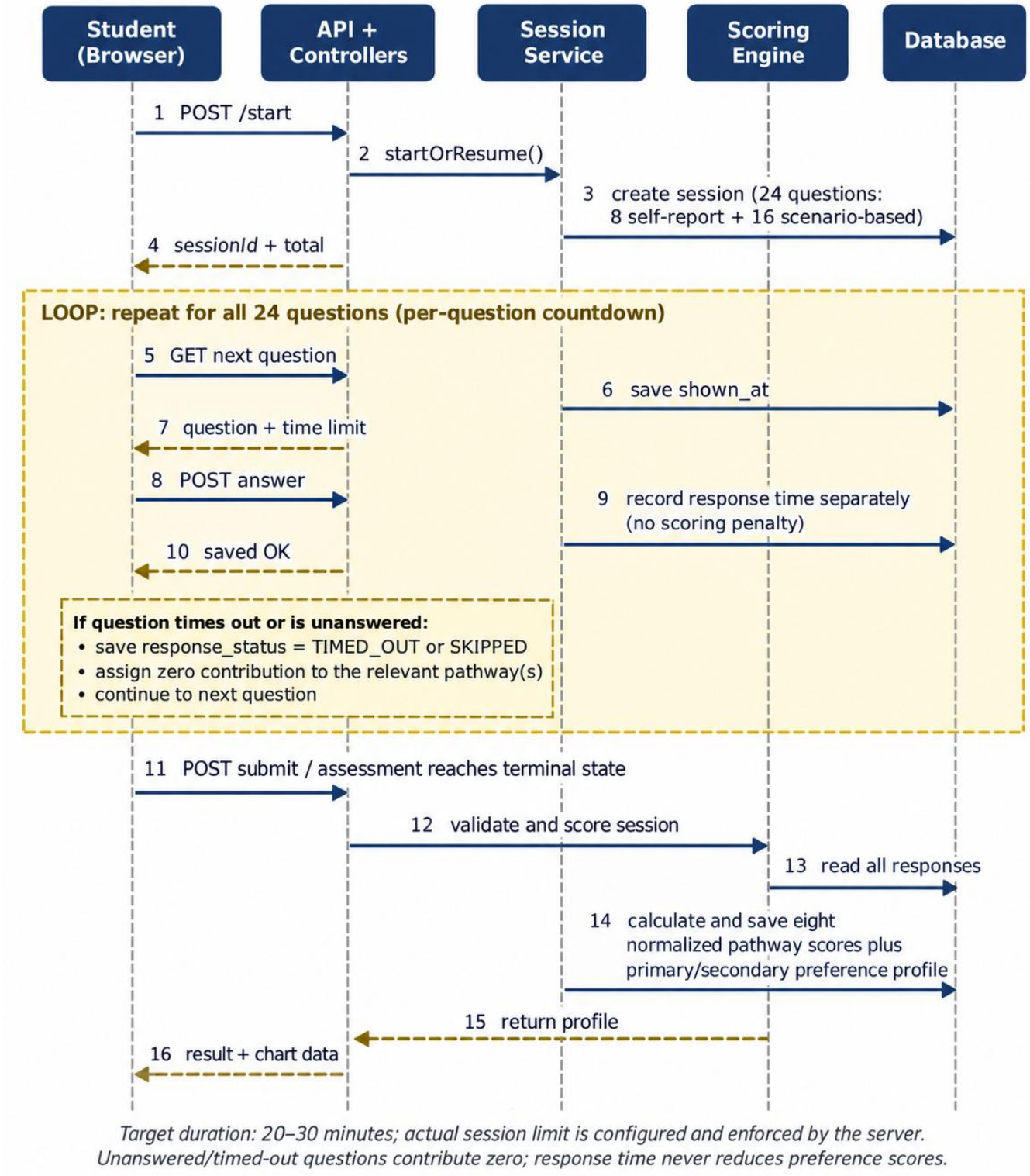


Figure 3: Sequence of one assessment of MindBridge

2.4 Data Design (Overview)

The data design contains the core tables plus scoring configuration. Details are in the LLD.

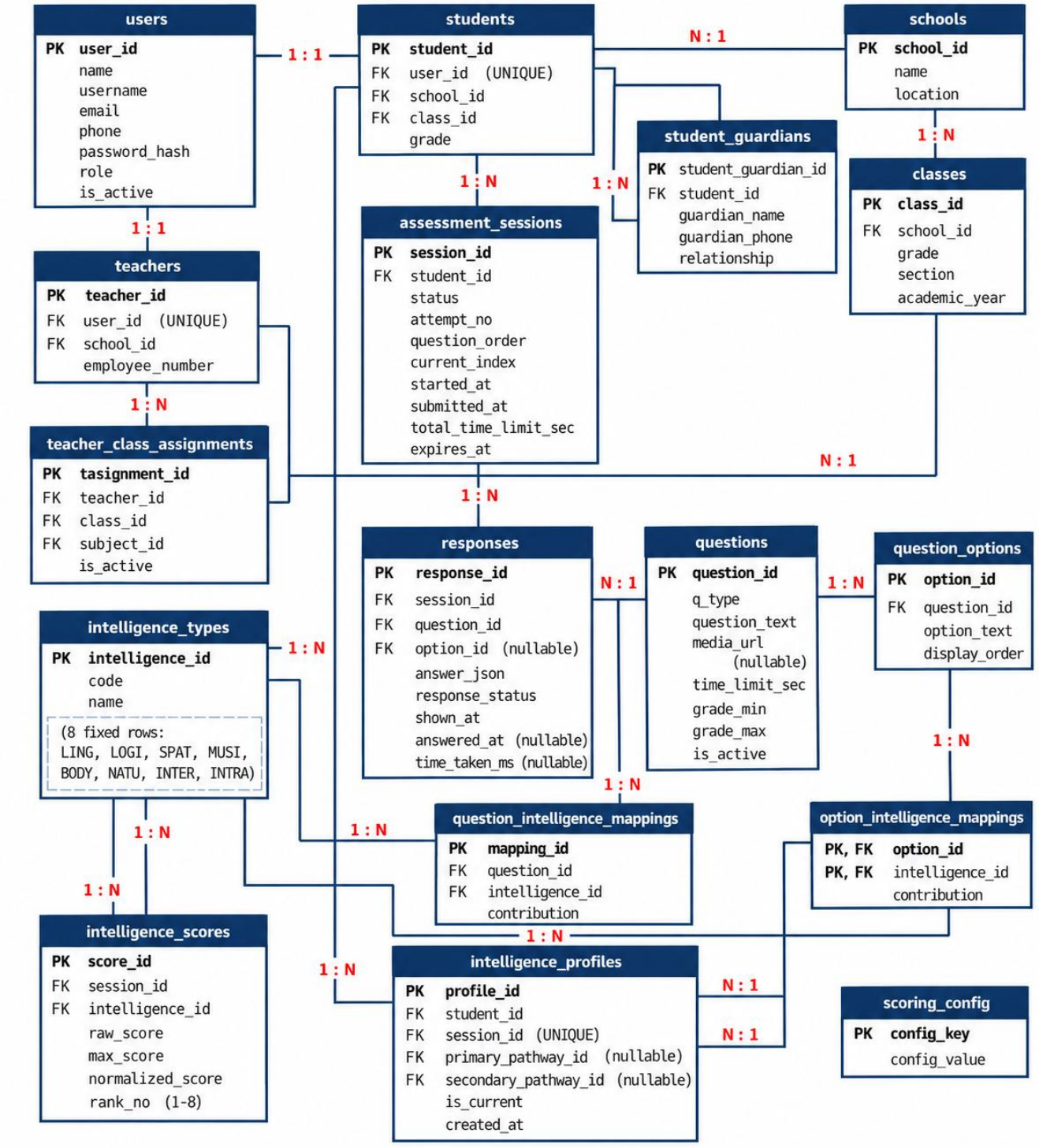


Figure 4: Entity relationship diagram of MindBridge

Only key columns are shown. Full columns are in Section 3.3. Questions and profile scores reference the eight approved learning pathways; profiles store primary and secondary preferences.

2.5 External Interfaces

Interface	Direction	Details
Student browser	In / Out	HTTPS REST calls using JSON
Module 2 (Content Learning)	Out	Authenticated profile API returns eight pathway preference scores and primary/secondary preferences.
Module 3 (Post-Assessment)	Out	Module 3 receives concept-level pre-test and post-test scores for learning-gain calculation; preference scores are not academic pre-test scores.
Media storage	In	Images and audio for learning content

Table 3: External Interfaces Table

2.6 Requirement to Design Mapping

Requirement	How the design meets it
PRE-F1, F2 Register, direct to questionnaire	Auth Service; UI redirects when no profile exists
PRE-F3 Question mapped to intelligence	questions.pathway_id or question_options.pathway_id; scenario options can map to preference pathways.
PRE-F4 Question types	Question Renderer with one component per type
PRE-F5 Record answer and time	responses table with shown_at, answered_at, time_taken_ms
PRE-F6, F7 Formula	Scoring Engine (Section 3.6)
PRE-F8 Rank, dominant, secondary	Profile Service ranks six pathways and stores primary/secondary preferences.
PRE-F9 Show chart	Result Page
PRE-F10 Observation	Optional observation service; separate from student preference score.
PRE-F12 Admin edits questions	Admin Question Manager and Question Service
PRE-F13 Re-take	assessment_sessions.attempt_no; only latest profile is_current
PRE-N6 Duration 15 minutes	20 questions; per-question limits add up to 900 seconds; total limit enforced on server
PRE-N3 Timer not reset	Server stores shown_at; client reads remaining time from server
PRE-N4 Save on drop	Every answer saved on submit; session resumes at current_index
PRE-N7 Security	JWT, role checks, hashed passwords, HTTPS

Table 4: Requirements to Design Mapping Table

3. Low Level Design (LLD)

3.1 Frontend Components

Component	Props / State	Behaviour
LoginPage	Form fields, validation errors	Calls the authentication API and redirects the user according to their role and assessment status.
RegisterPage	Form fields, validation errors	Registers a new student and redirects them to the instructions screen.
AssessmentPage	Session, current question, index	Controls the assessment flow: start, next question, save answer and submit.
InstructionScreen	None	Explains the assessment in simple language and shows instructions before starting.
ProgressBar	Current index, total questions	Shows progress through the 24-question assessment.
QuestionTimer	Question time limit, remaining time from server	Displays the countdown for the current question and the overall assessment; advances when a question times out.
QuestionRenderer	Question data, q_type	Selects the appropriate component based on SELF_REPORT or SCENARIO.
RatingtQuestion	Question text, scale 1–5, selected rating	Displays five rating options and returns the selected rating.
ScenarioQuestion	Question text, answer options, selected option	Displays scenario choices and returns the selected option_id.
ResultPage	Preference profile, pathway scores, rankings	Displays all eight indicative preference scores and highlights primary and secondary preferences.
ObservationForm (optional)	Student ID, eight pathway ratings, remarks	Allows authorized teachers or parents to record observations separately from student preference scores, if this feature is enabled.

Table 5: Frontend Components Table

3.2 Backend Classes

Class	Main methods	Notes
AuthController / AuthService	register(), login(), verifyToken()	bcrypt hashing; JWT expiry 2 hours
AssessmentController	start(), getNext(), saveAnswer(), submit()	Thin layer; calls SessionService
SessionService	startOrResume(), nextQuestion(), recordAnswer(), timeoutAnswer(), finish()	Owns session state and timing
QuestionSelector	buildSet(grade)	Picks 24 questions; balances coverage across eight pathways and mixes order.
TimerService	markShown(), computeTime(), isExpired()	Server clock only; checks question limit and 20-30 minute total limit
ScoringEngine	scoreSession(sessionId)	Runs the preference-scoring steps in Section 3.6.
ItemEvaluator (interface)	evaluate(question, answer)	Returns response weight 0 to 1
RatingEvaluator	evaluate()	(rating - 1) / 4
ScenarioEvaluator	evaluate()	Maps the selected scenario option to a preference contribution; no correct/incorrect answer.
ProfileService	saveProfile(), getCurrent(studentId)	Marks old profile is_current = false
ObservationService	add(), getForStudent()	Stores optional observations separately; does not blend them into preference scores.
ConfigRepository	get(key)	Cached values for the scoring parameters
QuestionRepository, SessionRepository, ResponseRepository, ProfileRepository	CRUD	Only these classes access the database

Table 6: Backend Classes Table

3.3 Database Schema

1. users

Column	Type	Constraint	Purpose
user_id	INT	PK, auto increment	Uniquely identifies each user account.
name	VARCHAR(100)	NOT NULL	Stores the user's full name.
email	VARCHAR(254)	NULLABLE	Stores the email for communication, notifications and account recovery.
phone	VARCHAR(20)	NULLABLE	Stores the user's contact number.
password_hash	VARCHAR(255)	NOT NULL	Stores the securely hashed password, never the plaintext password.
role	ENUM(student, teacher, parent, admin)	NOT NULL	Determines the user's role and access permissions.
location	VARCHAR(100)	NULLABLE	City/Village of the user.

Table 7: Users Table

2. students

Column	Type	Constraint	Purpose
student_id	INT	PK, auto increment	Uniquely identifies each student profile.
user_id	INT	FK → users.user_id, UNIQUE, NOT NULL	Links the student profile to the corresponding user account.
school_id	INT	FK → schools.school_id, NOT NULL	Identifies the school in which the student is enrolled.
class_id	INT	FK → classes.class_id, NOT NULL	Identifies the student's current class or section.
admission_number	VARCHAR(50)	Optional; unique within a school	Stores the school's student admission identifier.

Table 8: Students Table

3. teachers

Column	Type	Constraint	Purpose
teacher_id	INT	PK, auto increment	Uniquely identifies each teacher profile.
user_id	INT	FK → users.user_id, UNIQUE, NOT NULL	Links the teacher profile to the corresponding user account.
school_id	INT	FK → schools.school_id, NOT NULL	Identifies the school where the teacher works.
employee_number	VARCHAR(50)	Optional; unique within a school	Stores the teacher's institutional employee identifier.

Table 9: Teachers Table

4. student_guardian:

Column	Type	Constraint	Purpose
guardian_id	INT	PK, auto increment	Uniquely identifies each guardian record.
student_id	INT	FK → students.student_id, NOT NULL	Links the guardian to the student.
guardian_name	VARCHAR(100)	NOT NULL	Stores the parent's or guardian's full name.
guardian_phone	VARCHAR(20)	NOT NULL	Stores the contact number for communicating about the student.
guardian_email	VARCHAR(254)	NULL	Stores the guardian's email address, if available.
relationship	VARCHAR(30)	NOT NULL	Records the relationship, such as mother, father or legal guardian.
is_primary	BOOLEAN	DEFAULT FALSE	Indicates the primary contact to use when contacting the family.

Table 10: Students' Guardian Table

5. schools

Column	Type	Constraint	Purpose
school_id	INT	PK, auto increment	Uniquely identifies each school.
school_name	VARCHAR(150)	NOT NULL	Stores the official school name.
address	TEXT	NULL	Stores the school's street address.
city	VARCHAR(100)	NULL	Stores the city or town where the school is located.
district	VARCHAR(100)	NULL	Stores the district for location and administrative purposes.
state	VARCHAR(100)	NULL	Stores the state in which the school is located.
pincode	VARCHAR(10)	NULL	Stores the postal code of the school.

Table 11: Schools Table

6. classes

Column	Type	Constraint	Purpose
class_id	INT	PK, auto increment	Uniquely identifies each class or section.
school_id	INT	FK → schools.school_id, NOT NULL	Identifies the school to which the class belongs.
grade	SMALLINT	CHECK between 6 and 8	Stores the grade level supported by the MVP.
section	VARCHAR(10)	NOT NULL	Identifies the section, such as A or B.
academic_year	VARCHAR(9)	NOT NULL	Identifies the academic year, such as 2026–2027.

Table 12: Classes Table

7. teacher_class_assignments

Column	Type	Constraint	Purpose
assignment_id	INT	PK, auto increment	Uniquely identifies each teaching assignment.
teacher_id	INT	FK → teachers.teacher_id, NOT NULL	Identifies the teacher assigned to the class.
class_id	INT	FK → classes.class_id, NOT NULL	Identifies the class the teacher handles.
subject_id	INT	FK → subjects.subject_id, NOT NULL	Identifies the subject taught by the teacher.

Table 13: Teacher-Class Assigments Table

*8.*intelligence_types (8 fixed rows)

Column	Type	Constraint	Purpose
intelligence_id	SMALLINT	PK	Uniquely identifies each learning intelligence type.
code	ENUM(LING, LOGI, VIS, MUSI, BODY, NATU, INTER, INTRA)	UNIQUE, NOT NULL	Stores the short code each intelligence type.
name	VARCHAR(30)	UNIQUE, NOT NULL	Stores the display name of the intelligence type.

Table 14: Intelligence Types Table

9.Questions

Column	Type	Constraint	Purpose
question_id	INT	PK, auto increment	Uniquely identifies each question.
q_type	ENUM(SELF_REPORT, SCENARIO)	NOT NULL	Identifies whether the question is self-report or scenario-based.
question_text	TEXT	NOT NULL	Stores the question displayed to the student.
media_url	VARCHAR(255)	NULL	Stores an optional reference to an image or other supporting media, if needed.
time_limit_sec	SMALLINT	NOT NULL	Sets the maximum time allowed for answering the question.
grade_min	TINYINT	NOT NULL	Specifies the minimum grade for which the question is appropriate.
grade_max	TINYINT	NOT NULL	Specifies the maximum grade for which the question is appropriate.
is_active	BOOLEAN	DEFAULT TRUE	Indicates whether the question is available for assessment selection.

Table 15: Questions Table

10. question_intelligence_type_mappings

Column	Type	Constraint	Purpose
mapping_id	INT	PK, auto increment	Uniquely identifies each intelligence type mapping.
question_id	INT	FK → questions.question_id, NOT NULL	Identifies the question being mapped.
intelligence_id	SMALLINT	FK → intelligence_id, NOT NULL	Identifies the intelligence type receiving the contribution.
contribution	DECIMAL(4,3)	NOT NULL, CHECK 0–1	Defines the contribution associated with the mapping, where applicable.

Table 16: Question Intelligence Type Mapping Table

11.question_options

Column	Type	Constraint	Purpose
option_id	INT	PK, auto increment	Uniquely identifies each answer option.
question_id	INT	FK → questions.question_id, NOT NULL	Identifies the question to which the option belongs.
option_text	VARCHAR(255)	NOT NULL	Stores the answer option displayed to the student.
display_order	SMALLINT	NOT NULL	Specifies the order in which options appear.

Table 17: Question Options Table

12.Option_intelligence_type_mapping

Column	Type	Constraint	Purpose
option_id	INT	PK, FK → question_options.option_id	Identifies the option being mapped.
intelligence_id	SMALLINT	PK, FK → intelligence_id	Identifies the intelligence type receiving the contribution.
contribution	DECIMAL(4,3)	NOT NULL, CHECK between 0 and 1	Stores the preference contribution for that intelligence type.

Table 18: Option Intelligence Type Mapping Table

13.assessment_sessions

Column	Type	Constraint	Purpose
session_id	INT	PK, auto increment	Uniquely identifies an assessment session.
student_id	INT	FK → students.student_id, NOT NULL	Identifies the student taking the assessment.
status	ENUM(IN_PROGRESS, SUBMITTED, SCORED, EXPIRED)	NOT NULL	Tracks the current state of the assessment.
attempt_no	SMALLINT	NOT NULL	Records which attempt this is for the student.
question_order	JSONB	NOT NULL	Stores the ordered list of 24 question IDs selected for the session.
current_index	SMALLINT	NOT NULL, DEFAULT 0	Tracks the current question position for resuming the session.
started_at	TIMESTAMPTZ	NOT NULL	Records when the assessment started, using server time.
submitted_at	TIMESTAMPTZ	NULL	Records when the assessment was submitted.
total_time_limit_sec	INT	NOT NULL	Stores the configured overall session time limit.
expires_at	TIMESTAMPTZ	NOT NULL	Records the server-calculated session expiry time.

Table 19: Assessment Sessions Table

14.responses

Column	Type	Constraint	Purpose
response_id	BIGINT	PK, auto increment	Uniquely identifies each response record.
session_id	INT	FK → assessment_sessions.session_id, NOT NULL	Identifies the assessment session.
question_id	INT	FK → questions.question_id,	Identifies the question
		NOT NULL	being answered.
option_id	INT	FK → question_options.option_id, NULLABLE	Identifies the selected option; NULL for self-report answers.
answer_json	JSONB	NULL where an option ID suffices; otherwise stores the rating	Stores the student's answer, such as a 1–5 rating or selected option details.
shown_at	TIMESTAMPTZ	NOT NULL	Records when the question was shown, using server time.
answered_at	TIMESTAMPTZ	NULL	Records when the answer was received; NULL if unanswered or timed out.
time_taken_ms	INTEGER	NULL	Records the time taken to respond, in milliseconds.
timed_out	BOOLEAN	NOT NULL, DEFAULT FALSE	Indicates whether the question's time limit expired.

Table 20: Responses Table

15.intelligence_scores (UNIQUE session_id + intelligence_id)

Column	Type	Constraint	Purpose
score_id	INT	PK, auto increment	Uniquely identifies an intelligence type score record.
session_id	INT	FK → assessment_sessions.session_id, NOT NULL	Identifies the assessment session for which the score was calculated.
intelligence_id	SMALLINT	FK → intelligence_id, NOT NULL	Identifies the learning-preference intelligence type.
raw_score	DECIMAL(7,3)	NOT NULL	Stores the sum of the student's preference contributions for this Intelligence type.
max_score	DECIMAL(7,3)	NOT NULL	Stores the maximum possible contribution for this intelligence type based on the applicable questions and mappings.
profile_value	DECIMAL(5,2)	NULL if unavailable	Stores the normalized preference score from 0 to 100.
rank_no	SMALLINT	NULL if unranked	Stores the intelligence type's ranking from 1 to 8.

Table 21: Intelligence Scores Table

16.intelligence_profile

Column	Type	Constraint	Purpose
profile_id	INT	PK, auto increment	Uniquely identifies each learning-preference profile.
student_id	INT	FK → students.student_id, NOT NULL	Identifies the student to whom the profile belongs.
session_id	INT	FK → assessment_sessions.session_id, UNIQUE, NOT NULL	Identifies the assessment session that generated the profile.
primary_pathway_id	SMALLINT	FK → pathway_types.intelligence_id, NULL if unavailable	Identifies the student's primary learning preference.
secondary_pathway_id	SMALLINT	FK → pathway_types.intelligence_id, NULL if unavailable	Identifies the student's secondary learning preference.
is_current	BOOLEAN	NOT NULL, DEFAULT TRUE	Indicates whether this is the student's current profile.
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Records when the profile was created.

Table 22: Intelligence Profile Table

17.Observations

Column	Type	Constraint	Purpose
observation_id	INT	PK, auto increment	Uniquely identifies an observation record.
student_id	INT	FK → students.student_id, NOT NULL	Identifies the student being observed.
observer_user_id	INT	FK → users.user_id, NOT NULL	Identifies the teacher or parent who submitted the observation.
observer_role	VARCHAR(20)	NOT NULL	Records the observer's role at submission time.
remarks	TEXT	NULL	Stores optional descriptive observations.
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Records when the observation was submitted.

Table 23: Observations Table

18.observation_ratings

Column	Type	Constraint	Purpose
observation_id	INT	FK → observations.observation_id, NOT NULL	Identifies the observation to which the rating belongs.
intelligence_id	SMALLINT	FK → intelligence_id, NOT NULL	Identifies the learning-preference intelligence type being observed.
rating	SMALLINT	NOT NULL, CHECK between 1 and 5	Stores the observer's rating for that pathway.

Table 24: Observation Ratings Table

*19.*scoring_config

Config key	Type	Default	Meaning
near_tie_margin	DECIMAL(5,2)	Configurable after pilot	Maximum score difference, in percentage
			points, for displaying two leading intelligence types as similar preferences.
rating_max	SMALLINT	5	Highest rating on the self-report scale (1–5).
question_count	SMALLINT	24	Total questions per assessment session: 8 self-report and 16 scenario-based.
self_report_count	SMALLINT	8	Number of self-report questions per session.
scenario_count	SMALLINT	16	Number of scenario-based questions per session.
target_time_min_sec	INT	1200	Lower end of the target assessment duration: 20 minutes.
target_time_max_sec	INT	1800	Upper end of the target assessment duration: 30 minutes.
min_answered	SMALLINT	18	Minimum answered questions required to save a profile, if you retain the existing 75% completion policy.

Table 25: Scoring Configurations Table

3.4 API Design

All endpoints use HTTPS and JSON. Authentication uses a bearer JWT. Student endpoints are scoped to the authenticated student; teacher access is limited to students in assigned classes; administrator endpoints require the ADMIN role. Public registration creates student accounts only. Teacher and administrator accounts are provisioned by an administrator.

Method	Endpoint	Access	Contract
POST	/api/auth/register	Public	Create a student user and student profile in one transaction.
POST	/api/auth/login	Public	Verify credentials and return a short-lived JWT containing user ID and role.
POST	/api/assessment/start	Student	Resume the student's active session or create a new eligible attempt with a fixed 24-question order and snapshotted session time limit.
GET	/api/assessment/{sessionId}/next	Session owner	Return the current question, safe answer choices or rating scale, server-derived remaining time, and progress. Never return pathway mappings or scoring contributions.
POST	/api/assessment/{sessionId}/answer	Session owner	Save the answer for the current question once, using server timestamps. Return the next position and save status.
POST	/api/assessment/{sessionId}/submit	Session owner	Close the session, calculate eligible scores, and return either the profile or an insufficient-response result.
GET	/api/students/{studentId}/profile	Self, assigned teacher, admin	Return the current profile and all eight pathway scores. Enforce ownership or class assignment in the backend.
POST	/api/students/{studentId}/observations	Assigned teacher	Save supplementary observation ratings separately from student preference scores.
GET, POST	/api/admin/questions	Admin	List and create questions and their mappings. Only active, grade-eligible questions can be selected.
PUT, DELETE	/api/admin/questions/{questionId}	Admin	Update or deactivate a question. If already used in a response, do not change its text or scoring mappings in place; create a replacement question and deactivate the old one.
GET, PUT	/api/admin/config	Admin	Read or update assessment configuration. Validate counts, time range, Rating maximum, completion threshold, and near-tie margin.

Table 26: API Design Table

3.4.1 Assessment API payloads

GET /api/assessment/{sessionId}/next returns only the content needed to render the current item. For a self-report item, the answer control is the configured 1-to-5 scale. For a scenario, the response contains the option IDs and display text, without intelligence type mappings.

Example response:

Field	Example value
sessionId, index, total	"s-123", 5, 24
question	{ "id": "q-21", "type": "SELF_REPORT", "prompt": "I remember diagrams better than paragraphs.", "ratingScale": [1, 2, 3, 4, 5], "timeLimitSec": 45 }
remainingSessionSec	1260 (calculated by the server)

Table 27: Example Response Table

POST /api/assessment/{sessionId}/answer accepts exactly one answer shape: {"questionId":"q-21","answer":{"rating":4}} for self-report, or {"questionId":"q-22","answer":{"optionId":"opt-3"}} for scenario. The backend verifies the question is current, the answer shape matches the question type, and the selected option belongs to that question.

A successful submit returns status, answeredCount, requiredCount, profileAvailable, and (when available) the ranked eight pathway scores, primary and secondary preferences, nearTie, and the scoring configuration used. If the minimum completion threshold or pathway coverage is not met, return profileAvailable=false with reason=INSUFFICIENT_RESPONSES; do not fabricate zero scores for unavailable pathways.

3.5Question Bank Design

The first release contains 24 preference items: eight self-report items and sixteen scenario items. Interactive performance tasks are outside this assessment release. Task correctness must not be mixed into preference scores. The bank covers Grades 6–8 and the eight intelligence types listed in the updated schema.

Question type	Count	Authoring and scoring contract
Self-report	8	Use one age-appropriate preference statement for each pathway. Student selects a 1–5 rating; there is no correct answer.
Scenario	16	Present a learning situation with preference-based choices. Each option may map to one or more pathways with a 0–1 contribution; choices are not right or wrong.
Total	24	A session snapshots 24 eligible question IDs in a fixed order. Selection is limited by the student's grade and active question bank.

Table 28: Question Type Table

Intelligence type code	Intelligence type name
LING	Linguistic
LOGI	Logical-Mathematical
VIS	Visual
MUSI	Musical
BODY	Bodily-Kinesthetic
NATU	Naturalistic
INTER	Interpersonal
INTRA	Intrapersonal

Table 29: Intelligence Type Code and its name Equivalent Table

Question-bank release checks: all eight pathways must have scoring coverage; every scenario option must have at least one valid pathway mapping; each mapping contribution must be between 0 and 1; the active set must contain exactly eight self-report and sixteen scenario questions eligible for the student's grade. If the bank cannot produce a complete set, do not start the session; return a clear configuration error for the administrator.

Question text must be original, age-appropriate for Classes 6–8, and reviewed by a teacher before an administrator activates it. Keep pathway mappings server-side. Do not copy textbook or website question wording without permission.

3.6 Scoring Algorithm

3.6.1 Scoring rules

- The result represents a student-reported learning preference profile, not intelligence, ability, diagnosis, or proof that a pathway is the student's best way to learn.
- Response time is recorded for operational analysis only. It does not reduce or increase a preference score.
- Teacher observations are stored separately and do not change the student questionnaire score.
- There is no separate question-importance weight in the current schema. The pathway mapping contribution is the scoring weight for that response.

3.6.2 Formula

For a self-report rating r on the configured 1-to-5 scale, convert the rating to a response factor: f = (r - 1) / (max_rating - 1). For each pathway p mapped to that question, the response contribution is f × mappingContribution(p). A rating of 1 therefore contributes 0 and a rating of 5 contributes the full mapped contribution.

For a scenario response, the contribution to pathway p is the mapping contribution attached to the selected option for p. If the selected option has no mapping to p, the contribution is 0. Scenario choices are preference evidence and are never evaluated as correct or incorrect.

For each pathway p, rawScore(p) is the sum of its contributions across answered, non-timed-out items. maxScore(p) is the sum of the maximum possible contribution for p across those same answered items: the question mapping contribution for self-report, and the greatest option mapping contribution for p for each answered scenario. normalizedScore(p) = 100 × rawScore(p) / maxScore(p). Unanswered and timed-out items do not count as zero preference; they are excluded from both sums.

If maxScore(p) is zero for any pathway, that pathway is unavailable and the system does not publish a ranked profile for the attempt. A valid profile requires at least min_answered responses (18 of 24 by the current configuration) and scoring coverage for all eight pathways. Validate the 18-response threshold during the student pilot.

3.6.3 Ranking and near ties

Rank pathways by normalizedScore descending. Store all eight raw scores, maximum scores, normalized percentages, and ranks. The highest score is primary and the second-highest is secondary. If the difference between the top two percentages is at or below near_tie_margin, set nearTie=true and present them as similar preferences. Equal scores use the fixed pathway-code order only to make ordering deterministic; the order does not imply a meaningful preference difference. The near-tie margin is provisional and must be reviewed after the pilot.

Do not rescale the top pathway to 100% unless its normalized score is already 100%; each displayed percentage is an absolute normalized score against that pathway's available answered-item contribution.

3.7 Time Limit Design

Setting	MVP contract
Question count	24 total: 8 self-report and 16 scenario items.
Target duration	20–30 minutes. target_time_min_sec=1200 is the lower target; target_time_max_sec=1800 is the maximum session duration.
Session hard limit	Use 1800 seconds by default and store the effective total_time_limit_sec and expires_at on the session when it starts.
Question limit	Use each question's time_limit_sec. It is snapshotted by the question record used in the session; timer starts from server-recorded shown_at.
Timeout behavior	When a question limit expires, record a timed-out unanswered response, award no preference contribution for it, and advance. At the session hard limit, submit the session and mark remaining items unanswered/timed out.
Time scoring	No response-time penalty. Store shown_at, answered_at, and time_taken_ms using server time.

Table 30: Time Limit Design Table

The browser countdown is a display only. Refreshing or reconnecting must not reset shown_at, the question deadline, or the session deadline. The server decides whether a response arrived before the deadline; client clock values are ignored.

3.8 Session State and Timer Logic

State	Meaning	Allowed next state
IN_PROGRESS	Questions are being answered; fixed question order and current_index are saved.	SUBMITTED or EXPIRED
SUBMITTED	The student submitted, or the session hard limit ended. The server is closing input and evaluating completion.	SCORED when a valid profile is produced; remains submitted with no profile when completion is insufficient.
SCORED	The valid profile and eight pathway scores have been saved.	Terminal; a permitted retake creates a new session.
EXPIRED	The session was inactive for 24 hours and is no longer resumable.	Terminal; a permitted retake creates a new session.

Table 31: Session State and Timer Logic Table

- start(): return the existing IN_PROGRESS session for that student; otherwise create a new eligible attempt, select and store the fixed 24-question order, snapshot the session time limit, and set server timestamps.
- getNext(): return the current item. If it has already been shown, preserve the original shown_at and deadline. Never advance by trusting a client-supplied index.
- saveAnswer(): accept only the current question, validate answer type and ownership, calculate time from server timestamps, and persist once. A duplicate response returns 409 and does not overwrite the original answer.
- When a question deadline is reached, the next server request records a timeout and advances. When expires_at is reached, the server closes the session and marks remaining items unanswered. No background timer is required for correctness; every session endpoint checks server-side deadlines.
- After 24 hours without activity, mark the session EXPIRED. A student may resume only an IN_PROGRESS session; retake eligibility is controlled by the approved attempt policy. Every permitted retake uses a new attempt_no and preserves prior sessions and profiles.

3.9 Validation and Error Handling

Case	Server check	Response
Missing, invalid, or expired token	JWT authentication middleware	401 Unauthorized
Wrong role or teacher outside assigned class	Role and resource-level authorization	403 Forbidden
Unknown session, question, or option	Identifier exists and belongs to the requested resource	404 Not Found or 400 for malformed input
Answer does not match question type	Rating is an integer in 1..5, or option belongs to current scenario	400 Bad Request
Answer is not for the current item	Question ID matches the server current_index	409 Conflict
Duplicate answer	Unique session_id + question_id constraint	409 Conflict; existing answer is unchanged
Response after question/session deadline	Server timestamp and saved deadline	Record timeout; do not accept late answer; advance or submit
Submit below minimum or without pathway coverage	answeredCount and all eight denominators checked	200 with profileAvailable=false and reason=INSUFFICIENT_RESPONSES
No complete grade-eligible question set	Question-bank release validation	422 Unprocessable Entity; actionable admin configuration error
Database or unexpected service failure	Transaction rollback and server-side logging	500 Internal Server Error with a generic client message

Table 32: Validation and Error Handling Table

Persist each answer and its index advancement atomically. Scoring and profile creation must be transactional so the system cannot save a profile without all eight score rows or leave a session partially scored.

3.10 Security and Privacy

- Hash passwords with bcrypt using a suitable work factor. Never store or log plaintext passwords, JWT secrets, or database credentials.
- Require HTTPS in deployed environments. Sign short-lived JWTs with a strong server-side secret; keep secrets in environment settings, not source control.
- Enforce role and record-level access in backend services. Students can access only their own attempts and profiles. Teachers can access only students in their assigned classes. Admin endpoints require ADMIN.
- Do not expose pathway mappings, contributions, configuration secrets, or objective answer keys to student clients. The preference questionnaire has no correct answers.
- Validate all request bodies and use parameterized Prisma/database operations. Apply rate limits to authentication endpoints and avoid returning account-existence details unnecessarily.
- Collect only student data needed for the MVP. Define school consent, retention, and deletion procedures before real student data is used. Parent/guardian login and parent observations are not part of this role/API contract.

3.11 Acceptance Test Cases

ID	Scenario	Expected result
TC1	A new student registers and logs in.	Student account and student profile are created; student can start assessment.
TC2	A student starts, refreshes, and reconnects during an attempt.	The same session, question order, index, and server-derived remaining time resume.
TC3	Student requests a question mapping or another student's profile.	Mappings are never returned; unauthorized profile access returns 403.
TC4	Self-report rating is 0, 6, fractional, or missing.	Request is rejected with 400; no response is saved.
TC5	Scenario option belongs to a different question.	Request is rejected; no response or index change is committed.
TC6	Student submits a second answer for the same current question.	409 Conflict; the first answer remains unchanged.
TC7	Question or total-session timer expires.	Server records timeout, response receives no preference contribution, and the session advances or submits.
TC8	Student changes the browser clock.	Server deadlines and result are unaffected.
TC9	Student answers fewer than 18 questions.	No profile is saved; API returns INSUFFICIENT_RESPONSES and the retake action.
TC10	A pathway has no eligible answered-item denominator.	No ranked profile is published; the result identifies insufficient pathway coverage.
TC11	All eight pathway score rows are calculated.	Each normalized score is between 0 and 100 or explicitly unavailable; ranks are deterministic.
TC12	The two highest normalized scores are within near_tie_margin.	nearTie is true and the UI describes similar preferences without a fixed intelligence label.
TC13	A teacher reads a student profile outside the teacher's assignment.	403 Forbidden; no profile or observation data is disclosed.
TC14	An observation is submitted by an authorized teacher.	Observation and eight ratings are saved separately and do not alter the preference scores.
TC15	An administrator deactivates a question already used by a session.	Existing attempt and result remain reproducible; the question is excluded from future selections.

Table 33: Acceptance Test Cases Table